

RHEA SUDHEER

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Summary

Software engineer with an M.S. in Computer Science from Texas A&M and experience spanning systems programming, computer vision, machine learning, and production software. Built low-level C/C++ systems, real-time perception pipelines, multimodal ML systems, and production platforms supporting 100K+ records. Experienced engineering reliable software under latency, correctness, and deployment constraints and taking ambiguous technical problems from design through implementation, validation, and release.

Education

Texas A&M University — Master of Computer Science, GPA: 3.83/4.00

Aug 2024 – May 2026

PES University — B.Tech. Computer Science, GPA: 9.1/10

Dec 2020 – May 2024

Technical Skills

Languages: C++, Python, C, Java, SQL, PHP, JavaScript, Bash

ML/ Perception: PyTorch, TensorFlow, Scikit-learn, OpenCV, Computer Vision, Object Detection, CNNs, LSTMs, Transformers, RAG, FAISS

Systems: Linux, Multithreading, Synchronization, Virtual Memory, Paging, Scheduling, Concurrency, Distributed Systems, Networking

Backend / Infrastructure: REST APIs, MySQL, Flask, Laravel, Docker, Kubernetes, AWS, GCP, Kafka, Spark, CI/CD

Engineering Tools: Git, CMake, GDB/LLDB, QEMU/Bochs, Unit Testing, Profiling, Debugging, Code Review, System Reliability

Experience

Public Policy Research Institute, Texas A&M University

College Station, TX

Software Applications Developer I

Sep 2026 – Present

Student Research Assistant

Jan 2025 – Sep 2026

- Engineer and maintain a production PHP/Laravel + MySQL platform supporting 100,000+ public records; develop backend services, relational schemas, APIs, metadata-driven workflows, internal tools, dashboards, and reporting systems.
- Build reusable SQL and data-processing components for ingesting, transforming, joining, and validating multi-source data; automate detection of malformed, duplicated, incomplete, and inconsistent records before downstream use.
- Own features from ambiguous stakeholder requirements through design, implementation, testing, debugging, and release; diagnose application and database failures using logs, SQL investigation, targeted tests, and profiling and convert recurring failures into maintainable fixes and regression protection.
- Drive implementation across database schemas, backend logic, permissions, and user-facing workflows while coordinating changes across interconnected modules and preserving data integrity and existing application behavior.

Ernst & Young — Technology Consulting Intern

Bangalore, India | Jan 2024 – Jun 2024

- Evaluated enterprise Oracle/SAP systems across access control, change management, disaster recovery, security, and operational reliability; investigated exceptions with client technology teams and translated root causes into traceable remediation actions.

IBM — Machine Learning Technical Mentee

Bangalore, India | Dec 2022 – Mar 2023

- Built an end-to-end Python/Scikit-learn ML pipeline for Type-I diabetes prediction from heterogeneous multi-omic data, including validation, preprocessing, feature engineering, cross-validation, model selection, and inference; achieved 97.7% reported accuracy.

Thales — Artificial Intelligence Intern

Bangalore, India | Jun 2022 – Aug 2022

- Developed a real-time MobileNetV2 + SSD pedestrian-detection pipeline; profiled inference latency, model size, detection quality, and false/missed detections under deployment constraints.
- Applied SHAP, LIME, and saliency analysis to diagnose perception failures and validate model behavior, balancing latency, interpretability, and reliability in a safety-sensitive computer-vision workflow.

Projects

Custom x86 Operating System

C/C++, x86, QEMU/Bochs, GDB/LLDB

- Engineered core OS subsystems including demand-paged virtual memory, page-fault handling, physical-frame allocation, TLB-aware state management, timer-interrupt-driven preemptive scheduling, and safe thread lifecycle management.
- Implemented spinlocks/mutexes, inode-based storage, buffered I/O, and interrupt-driven non-blocking disk access; debugged memory, synchronization, scheduling, race-condition, and device-I/O failures across hardware/software boundaries.

Vision for the Blind: Real-Time Assistive Computer Vision

Python, TensorFlow, OpenCV, YOLOv3, LSTM

- Built an end-to-end assistive perception pipeline combining live camera input, YOLOv3 object/obstacle detection, temporal LSTM modeling across sequential frames, and audio feedback for visually impaired users.
- Integrated perception, temporal inference, and audio feedback into a real-time pipeline designed for low-latency obstacle awareness and temporally consistent, actionable guidance rather than isolated frame-level predictions.

Hybrid Self-RAG with External Knowledge Validation

Python, FAISS, LangChain, Transformers

- Built a multi-stage retrieval and verification system combining generation, claim extraction, live evidence validation, semantic scoring, conditional FAISS retrieval, and evidence-conditioned regeneration.
- Implemented threshold-driven fallback behavior that improved factual accuracy from 81% to 89%, reduced hallucination rate from 19% to 11%, and increased average FactScore from 0.65 to 0.87 while analyzing quality-latency trade-offs.

Multimodal Nutrition Intake Estimation from Vision and Sensor Data

Python, PyTorch, CNN, Time Series, Multimodal ML

- Developed a multimodal neural architecture combining CNN-encoded meal images, 288-step CGM time-series signals, and demographic/microbiome features through modality-specific encoders and learned feature fusion for caloric-intake estimation.
- Built preprocessing pipelines for heterogeneous sensor, image, and structured data, including missing-data handling, normalization, categorical encoding, and PCA-based dimensionality reduction while evaluating generalization and overfitting under limited-subject data.

Publications

Revolutionizing Healthcare: A Review Unveiling the Transformative Power of Digital Twins

Enhancing Robustness in Audio Visual Speech Recognition: Preprocessing Approach with Transformer and CTC Loss

Leukaemia: A Comparative Analysis of Deep Learning Models Using ALL Dataset